MA 301

Final

Spring - 2013

Problem	points	out of
1		10
2		10
3		10
4		10
5		10
6		14
7		15
8		18
9		13
Score		100

Name: ___________________

HR: ___________________

Signature: ___________________

READ BEFORE YOU START
* Show all work.
* Change your calculator mode to RADIANS.
* Sloppy solutions may be penalized.
* Use calculators ONLY for computations.
* You must write up every problem in a clear and organized way.
* If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Given $\mathbf{F} = [2, 2y + z, y + 1]$

(a) Show that $\mathbf{F}$ is conservative in two different ways

method 1	method 2

(b) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is a straight line from $(-1, 0, 2)$ to $(0, 2, -6)$

Problem 2: Let $\mathbf{F} = [y, x\sin y]$ and C be the boundary (with positive orientation) of the region between $y = \frac{\pi}{2}x,\ \ y = \frac{\pi}{2}$ and $0 \le x \le 1$ in the xy-plane.

Sketch C and compute $\oint\limits_C \mathbf{F} \cdot d\mathbf{r}$

Problem 3: Let $\mathbf{F} = [x, ye^z, e^z]$ and S be the surface (with outward orientation) of the box $|x| \le 1$, $|y| \le 2$, $0 \le z \le 1$.

Compute $\displaystyle\iint_S \mathbf{F} \cdot \mathbf{n}\, dA$

Problem 4: Let $f(x) = \begin{cases} 0 & -2 < x < -1 \\ 1 + x & -1 < x < 0 \\ 1 - x & 0 < x < 1 \\ 0 & 1 < x < 2 \end{cases}$ and $f(x) = f(x+4)$

Find the Fourier Series of $f(x)$

Problem 5: Let $f(x) = \begin{cases} -x^2 & -1 < x < 0 \\ x^2 & 0 < x < 1 \end{cases}$ and $f(x) = f(x+2)$

(a) Graph $f(x)$ on the interval $[-3, 2]$

(b) Can we say that any of Fourier coefficients a_0, a_n, or b_n are zero? Why or why not?

(c) **Set-up**, but **DO NOT** compute the integral(s) for all the non-zero Fourier coefficients.
("$f(x)$" should not be part of the answer and the integrals should be simplified as much as possible)

Problem 6: Let $z_1 = -2 - 2i, \; z_2 = i.$ Answer the following:

(a) $z_1 z_2 =$

(b) $z_1 - z_1 z_2 =$

(c) $|z_1 z_2| =$

(d) Convert z_1 to Euler's Polar form:

(e) Convert z_2 to Euler's Polar form:

(f) (Polar Form) $(z_2)^{76} =$

(g) (Polar Form) Solve for z: $\quad z_2 z^{1/3} = z_1$

Problem 7: Determine the convergence of each series. If convergent, find the sum. Justify all work.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{n-1}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-6)^n}{5^{n+1}}$

(c) $\displaystyle\sum_{n=1}^{\infty} \sin\left(\frac{\pi n}{2n-1}\right)$

(d) $\displaystyle\sum_{n=0}^{\infty} \frac{3^{2n}}{10^n}$

(e) $\displaystyle\sum_{n=4}^{\infty} \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$

Problem 8: Answer the following true/false questions

T / F	The line integral of any vector field $\mathbf{F}$ around any closed curve C is zero
T / F	If $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for one particular closed path, the $\mathbf{F}$ is path-independent.
T / F	If $\mathbf{F}$ is path-independent, then there is a potential function for $\mathbf{F}$
T / F	If $\int_C \mathbf{F} \cdot d\mathbf{r} \neq 0$ for some closed path C, then $\mathbf{F}$ is path-dependent.
T / F	If $\mathbf{F} = \nabla f$ then $\mathbf{F}$ is conservative.
T / F	If $\mathbf{F}$ is path-independent and C is any closed curve, then $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$
T / F	If S_1 is a rectangle with area = 1 and S_2 is a rectangle with area = 2, then $2 \iint_{S_1} \mathbf{F} \cdot d\mathbf{A} = \iint_{S_2} \mathbf{F} \cdot d\mathbf{A}$
T / F	Suppose $\mathbf{F} = [1, 1, 1]$ and S is a closed surface with positive orientation. Then $\iint_S \mathbf{F} \cdot \mathbf{n}\, dA > 0$.
T / F	Let S be a sphere of radius 1 centered at the origin and oriented outward. If $\iint_S \mathbf{F} \cdot d\mathbf{S} = 0$, then $\mathbf{F} = \mathbf{0}$
T / F	Let S be the surface of a cube centered at the origin with side length 2 and oriented outward. If $\mathbf{F} = 2\mathbf{i} - \mathbf{k}$ then the flux of $\mathbf{F}$ through S is 27.
T / F	If $\mathbf{F}$ is a vector-field such that div$\mathbf{F} = 3$ and S is a closed surface oriented outward, then $\iint_S \mathbf{F} \cdot d\mathbf{S}$ is equal to one-third the volume enclosed by S.
T / F	Suppose $f(x) = f(x+6)$ for all x, then we say f is periodic with period 3.
T / F	$\sqrt{-5} \cdot \sqrt{-5} = 5$.
T / F	If $\sum_{n=1}^{\infty} a_n$ converges, then so does $\sum_{n=100}^{\infty} a_n$.
T / F	If $\sum_{n=1}^{\infty} a_n$ diverges, then so does $\sum_{n=100}^{\infty} a_n$.
T / F	If $\sum_{n=10}^{\infty} a_n$ converges, then $\lim_{n\to\infty} \cos(a_n) = 1$.
T / F	If $\sum_{n=1}^{\infty} a_n$ converges, then so does $\sum_{n=1}^{\infty} \frac{1}{(a_n)^2}$.
T / F	If $\sum_{n=1}^{\infty} a_n$ converges, then it is possible for the partial sums to converge to $\frac{3}{2}$.

Problem 9: Answer the following questions

(a) Let $\mathbf{F}$ vector field such that curl $\mathbf{F} = \mathbf{0}$.
Match each expression in the left column with the most appropriate term in the right column.

$\int_C \mathbf{F} \cdot d\mathbf{r}$	conservative
$\oint_C \mathbf{F} \cdot d\mathbf{r}$	path independent
$\mathbf{F}$	zero

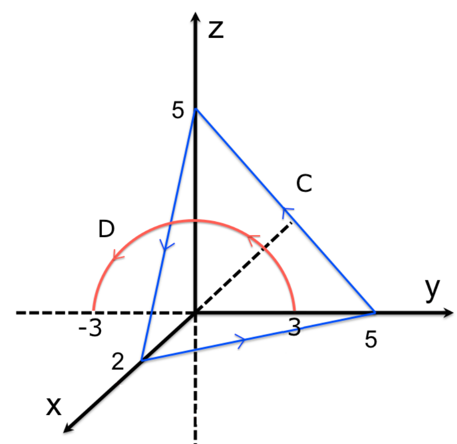
(b) Match each the Theorem in the left column with the most appropriate statement in the right column.

Divergence Theorem	The surface of a full sphere of radius 4
Stokes' Theorem	The surface of a half-sphere of radius 4
Green's Theorem	A disk of radius 4

(c) Let $\mathbf{F} = \nabla\left(\dfrac{x^2}{y+z+1}\right)$ and let C and D be shown in the figure. Compute

$$\int_C \mathbf{F} \cdot d\mathbf{r} =$$

$$\int_D \mathbf{F} \cdot d\mathbf{r} =$$



(d) Suppose $\displaystyle\sum_{k=1}^{\infty} \left(\frac{c}{5}\right)^k = \frac{c}{6}.$ Find all values of c which satisfy this expression.